

Learning-based slip detection for adaptive grasp control in robotic manipulation

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Institut Cea-List, Nano-INNOV, Palaiseau

Composition du jury

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- Liming CHEN – Rapporteur
Professeur des universités, Ecole Centrale de Lyon
- Jean-Pierre GAZEAU – Examineur
Ingénieur de recherche CNRS, Institut P'
- Richard BEAREE – Examineur
Professeur des universités, ENSAM Lille
- Maria MAKAROV – Examinatrice
Enseignant-chercheur, CentraleSupélec - Université Paris-Saclay

- Mathieu GROSSARD – Directeur de thèse
Directeur de recherche, CEA-List
- Saifeddine Aloui – Co-encadrant
Ingénieur de recherche, CEA-Leti

From pre-programmed pick-and-place to contact-rich manipulation

Adaptation / Versatility
↓

1. Pre-programmed pick-and-place

- Hard-coded motions in **controlled environments**



HRV Magnetics

From pre-programmed pick-and-place to contact-rich manipulation

Adaptation / Versatility
↓

1. Pre-programmed pick-and-place
2. **Force-controlled grasping**
 - Deformable and fragile objects



qb SoftClaw

From pre-programmed pick-and-place to contact-rich manipulation

- Adaptation / Versatility ↓
1. Pre-programmed pick-and-place
 2. Force-controlled automation
 3. Force-controlled manipulation
 4. **Dexterous, contact-rich manipulation**
 - Adaptation to **unstructured environments**
 - Multi-contact planning for in-hand manipulation
 - Robust handling of unknown, irregular, and deformable objects



TRACEBOT

Need for dexterity

Logistics

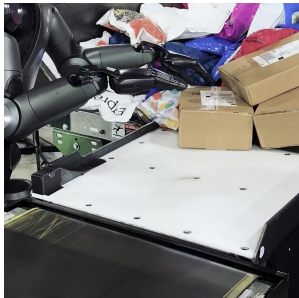
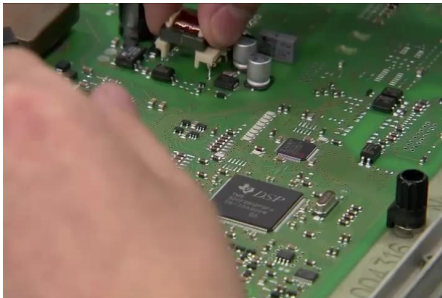


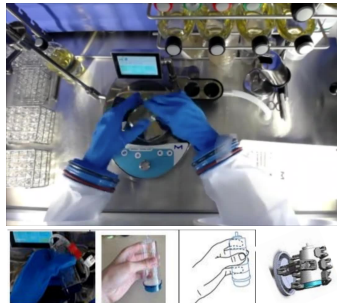
Figure A1

Manufacturing



Z-AXIS Inc.

Critical environments



EU project TraceBot

Roadmaps: EU robotics (SPARC/ADRA) call for **adaptive, tactile-aware** manipulation.

Enablers of dexterity 1/3

Versatile grippers

Enablers of dexterity 1/3

Versatile grippers

- Multi-digit grippers, articulated fingers

Increasing degrees of freedom (DoF) →



Robotiq



RBO hand TU Berlin



Shadow Robot

Enablers of dexterity 1/3

Versatile grippers

- Multi-digit grippers, articulated fingers
- Reconfigurability for human grasps

Increasing degrees of freedom (DoF) →



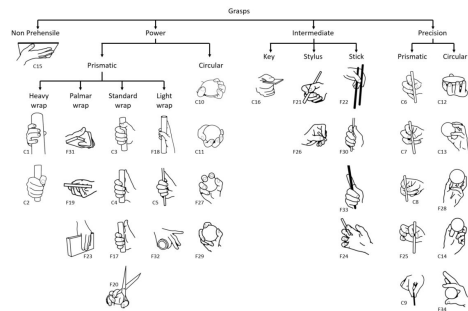
Robotiq



RBO hand TU Berlin



Shadow Robot



Escorcia-Hernández et al., 2023

Enablers of dexterity 1/3

Versatile grippers

- Multi-digit grippers, articulated fingers
- Reconfigurability for human grasps
- Controlled contact forces

Increasing degrees of freedom (DoF) →



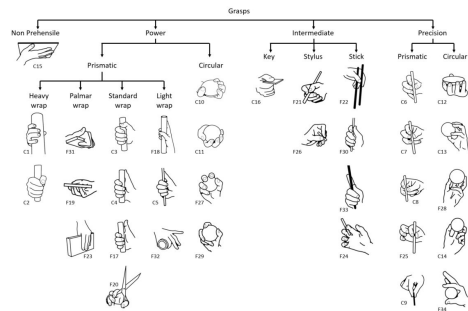
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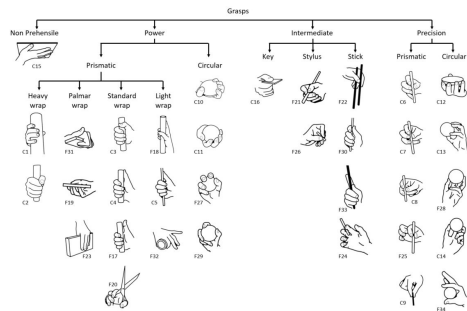
Robotiq



RBO hand TU Berlin



Shadow Robot



Escorcia-Hernández et al., 2023

Contribution

- ✓ Ensuring **stability** of multifingered grasps
- ✓ **Coordination** of contact forces

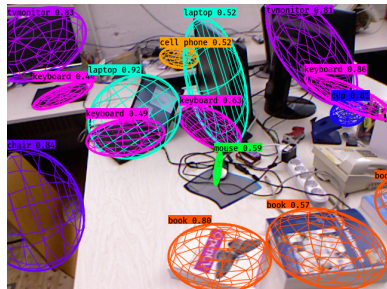
Enablers of dexterity 2/3

Perception for adaptation

Enablers of dexterity 2/3

Perception for adaptation

- Vision for global scene understanding

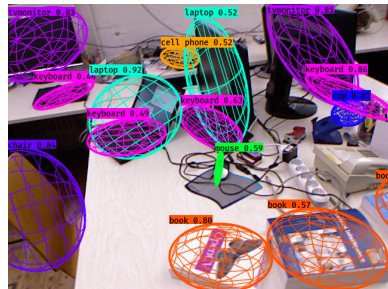


QUT center for robotics TUM-A2

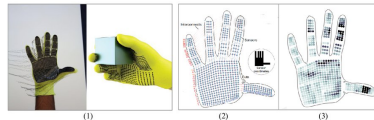
Enablers of dexterity 2/3

Perception for adaptation

- **Vision** for global scene understanding
- **Tactile** for local contact state



QUT center for robotics TUM-A2

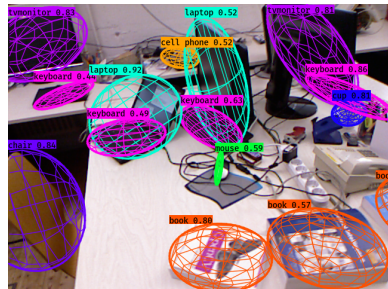


Sundaram et al., 2019

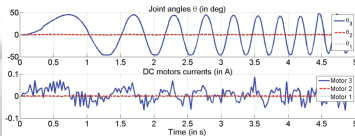
Enablers of dexterity 2/3

Perception for adaptation

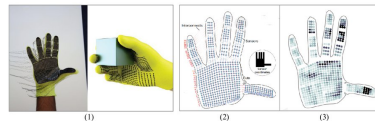
- Vision for global scene understanding
- Tactile for local contact state
- Proprioception for internal state



QUT center for robotics TUM-A2



CEA Hand, Martin and Grossard, 2014
Garrec, P. 2008



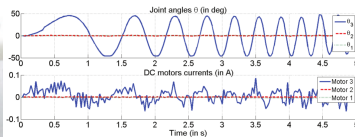
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Enablers of dexterity 2/3

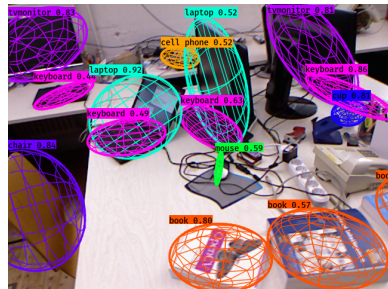
Perception for adaptation

- Vision for global scene understanding
- Tactile for local contact state
- Proprioception for internal state

→ Closing feedback loops



CEA Hand, Martin and Grossard, 2014
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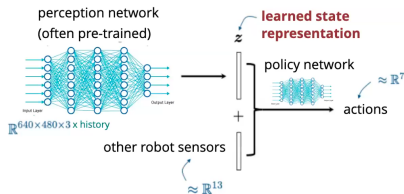
QUT center for robotics TUM-A2

Contribution

- ✓ Tactile perception for object slip detection
- ✓ Feedback loop with **adaptive** force modulation

Enablers of dexterity 3/3

Learning robust policies

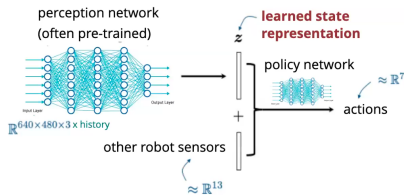


Russ Tedrake, 2023

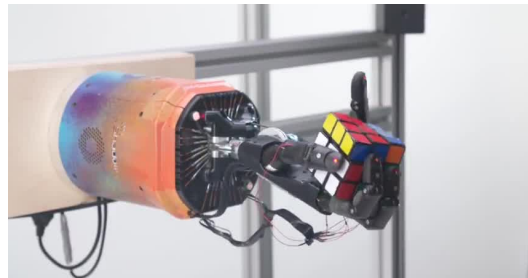
Enablers of dexterity 3/3

Learning robust policies

- High-DoF gripper control



Russ Tedrake, 2023

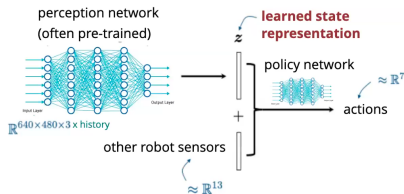


OpenAI et al., 2019

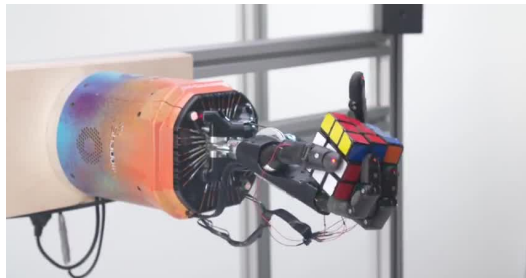
Enablers of dexterity 3/3

Learning robust policies

- High-DoF gripper control
- Multimodal sensor fusion



Russ Tedrake, 2023

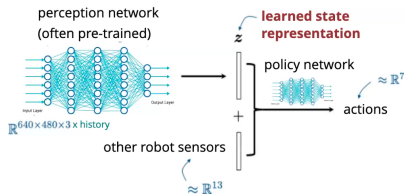


OpenAI et al., 2019

Enablers of dexterity 3/3

Learning robust policies

- High-DoF gripper control
- Multimodal sensor fusion
- Implicit models



Russ Tedrake, 2023

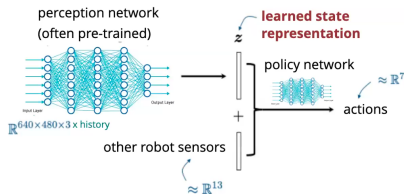


Toyota Research Institute

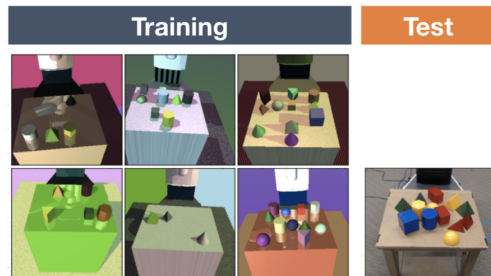
Enablers of dexterity 3/3

Learning robust policies

- High-DoF gripper control
- Multimodal sensor fusion
- Implicit models
- Data-driven **robustness** and **generalization**



Russ Tedrake, 2023

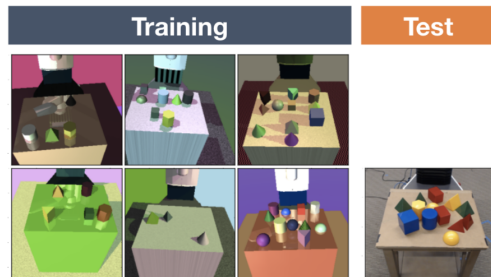
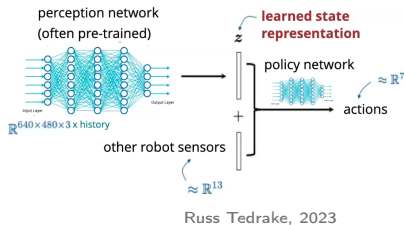


Tobin et al., 2017

Enablers of dexterity 3/3

Learning robust policies

- High-DoF gripper control
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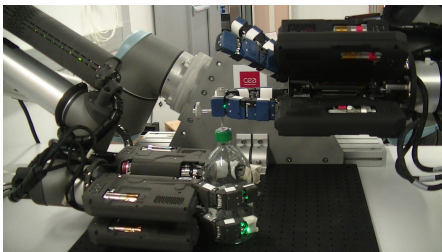


Tobin et al., 2017

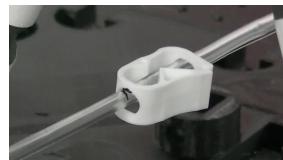
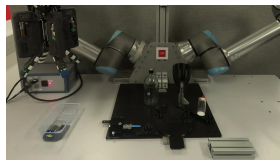
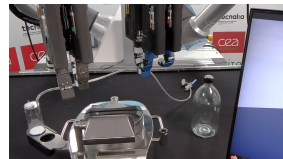
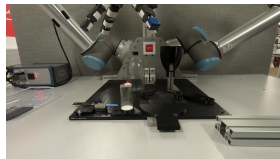
Contribution

- ✓ Learning **robust tactile perception**
- ✓ Data-driven **multimodal sensor fusion**

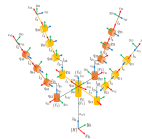
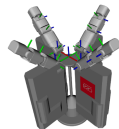
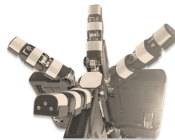
Experimental setup



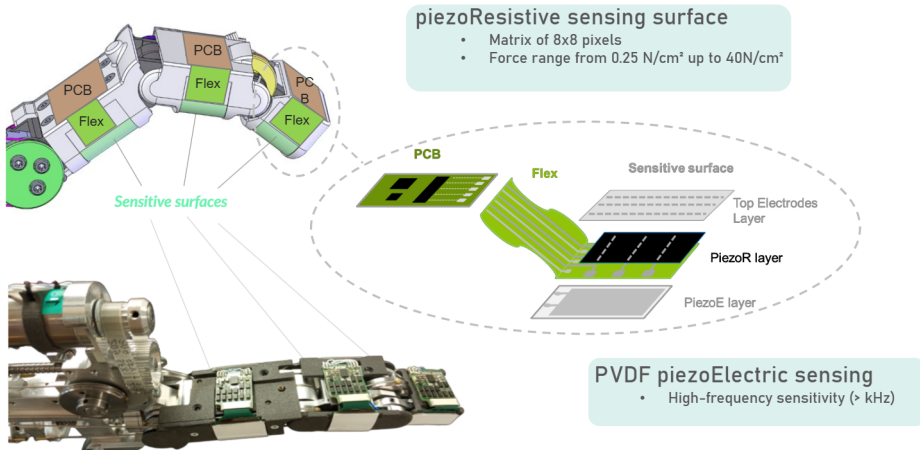
EU project TraceBot



- Four-fingered 14-DoF gripper
- Haptic sensing capabilities
- Hybrid tactile pads on each phalanx and palm

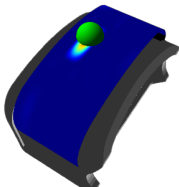
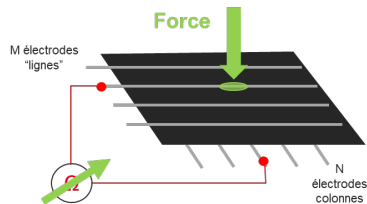
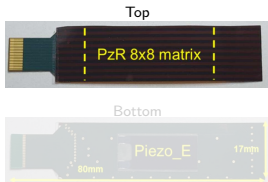


Hybrid tactile sensor PzE / PzR



Piezoresistive sensor (PzR)

CEA thin flex-based sensor



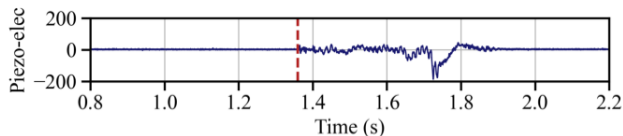
Piezoresistivity

Electrical **resistance** decreases locally under mechanical **pressure**.

- 100 μm piezoresistive *VelostatTM* polymer
 - Screen-printed silver ink electrodes
 - Two orthogonal sheets for 8×8 taxel matrix (lines $\times$ columns)
- **spatial distribution of pressure, contact points**

Piezoelectric sensor (PzE)

CEA thin flex-based sensor



Piezoelectricity

Mechanical stress produces electric charge in the material.

- PVDF-TrFE transducers between *PEDOT:PSS* electrodes
 - 10kHz sampling, effective bandwidth 30 Hz–2.5 kHz.
- detects high-frequency events and vibrations

